

J. David Muñoz-Bolaños

R&D Optical Engineer | Optical Metrology Specialist

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PROFESSIONAL SUMMARY

R&D Optical engineer based in **Innsbruck** with a magic combination of **Industrial Machine Vision** expertise and **High-Precision Optical Metrology**. Proven track record in developing illumination concepts, algorithmics (Python), and hardware integration for automated inspection systems. Experience bridging the gap between "code" and "hardware," from implementing inspection systems on high-speed industrial lines to correcting optical aberrations using **MEMS devices**. Eager to apply expertise in **optics, vision, and algorithms** to TriLite's next-generation Laser Beam Scanning (LBS) systems.

TECHNICAL COMPETENCIES

Vision Technology & Algorithmics

MVTec HALCON (Industrial software), Image Processing Pipelines, **Python**, GPU Acceleration (CuPy) for real-time processing, **Laser Profilometry**, Camera/Laser Calibration.

Optics & Lab Management

Illumination/Optical Design, **Zemax OpticStudio** (Simulation & Tolerance Analysis), Adaptive Optics (**Texas Instruments DMD**, SLM), Interferometry, **Laser Beam Shaping**, System Integration, Prototyping.

Hardware Control

FPGA Real-time control (LabVIEW), Motion Control, High-speed Detectors (PMT/APD), Semiconductor-based **MEMS integration**.

Languages

English (Proficient), German (Intermediate/Working Proficiency), Spanish (Native).

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

PhD Researcher - Optical Systems Engineering

Innsbruck, Austria

Jan 2022 – Present

- **High-Speed Hardware Integration (MEMS):** Designed and built a control system for a **Texas Instruments DLP/DMD** to correct optical aberrations. This work mirrors laser beam scanning tasks, requiring high-speed/precision actuation and real-time feedback loops.
- **Algorithmics & GPU Acceleration:** Developed GPU-accelerated algorithms (Python/CuPy) to restore imaging quality in complex media. Optimized processing pipelines to handle large image stacks efficiently.
- **Optics Lab Management:** Responsible for the maintenance and improvement of the microscopy lab, including laser safety protocols, handling fine optical components, and aligning complex beam paths.

Leibniz Institute of Photonic Technology (IPHT)

R&D Assistant Researcher

Jena, Germany

Jun 2020 – Dec 2021

- **Prototyping:** Assembled multimodal imaging setups (Raman/Photothermal) from concept to functional prototype.
- **Hardware Integration:** Collaborated with scientific partners to qualify new detection methods for environmental microplastic analysis.

INTECOL S.A.S.

Machine Vision Engineer

Medellin, Colombia

Mar 2018 – Mar 2019

- **Industrial Deployment (OCR):** Led the full development cycle of a quality control system for bottling lines. Designed custom **LED illumination** concepts and programmed **HALCON** algorithms to reliably read and verify OCR codes on printed labels.
- **Safety-Critical Inspection:** Developed a **laser profilometry system** for the Metro de Medellin cable cars. Created the concept, integrated the vision hardware, and validated the system for outdoor environments.
- **Production Transfer:** Managed on-site installation, calibration, and provided technical support and training for operators.

EDUCATION

Medical University of Innsbruck

Ph.D. Advanced Microscopy & Optics (Available from March 2026)

Austria

Jan 2022 – Present

Friedrich Schiller University Jena

M.Sc. in Photonics

Germany

2019 – 2021

University of Cauca

B.Sc. in Physics Engineering

Colombia

2012 – 2018

SELECTED PUBLICATIONS & AWARDS

- J. D. Muñoz-Bolaños, et al. "**Scatter compensation** for enabling deep two-photon microscopy using a ****MEMS phase light modulator****," Proc. SPIE (2026).
- J. D. Muñoz-Bolaños, et al. "*****Wavefront corrections**** over large fields of view via beam cone tomography." Proc. SPIE (2025).
- **Awards:** ZEISS Summer School (2025), COLFUTURO Grant (2019).

REFERENCES

Prof. Monika Ritsch-Marte

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Medical University of Innsbruck

Prof. Jürgen Popp

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Leibniz Institute of Photonic Technology